

Optimization for Learning Exercises

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Linear algebra

Exercise 1 Verify the Schur complement formula.

Derivatives with respect to matrices

Exercise 2 Let $X : \mathbf{R} \rightarrow \mathbf{R}^{m \times n}$ and $Y : \mathbf{R} \rightarrow \mathbf{R}^{n \times p}$. Show that

$$\frac{d}{dt}(X(t)Y(t)) = \frac{dX(t)}{dt}Y(t) + X(t)\frac{dY(t)}{dt}$$

Answer of Exercise 2 Let $F(t) = X(t)Y(t)$. With Matlab-like notation $F_{i,j}(t) = X_{i,:}(t)Y_{:,j}(t)$, and hence

$$\frac{dF_{i,j}(t)}{dt} = \frac{dX_{i,:}(t)}{dt}Y_{:,j}(t) + X_{i,:}(t)\frac{dY_{:,j}(t)}{dt}$$

from which the result follows.

Exercise 3 Show that

$$\frac{\partial}{\partial X} \text{tr} AX = A^T$$

where $X \in \mathbf{R}^{m \times n}$ and $A \in \mathbf{R}^{n \times m}$

Answer of Exercise 3 We have $\text{tr} AX = (\text{vec} A^T)^T \text{vec} X$, from which the result follows.

Exercise 4 Show that

$$\frac{\partial}{\partial X} \ln \det X = X^{-T}$$

where $X \in \mathbf{R}^{m \times m}$ and $\det X > 0$. Hint: Remember Jacobi's formula.

Answer of Exercise 4 It holds that

$$\frac{\partial}{\partial X_{i,j}} \ln \det X = \frac{1}{\det X} \frac{\partial X}{\partial X_{i,j}} = (\mathbf{adj} X)_{j,i}$$

where the last equality follows from Jacobi's formula. The result then follows from the well-known formula

$$X^{-1} = \frac{\mathbf{adj} X}{\det X}$$

Exercise 5 Show that

$$\frac{\partial}{\partial X} a^T X^{-1} b = -X^{-T} a b^T X^{-T}$$

where $X \in \mathbf{R}^{m \times m}$ and $a, b \in \mathbf{R}^m$. Hint: First use the identity $XX^{-1} = I$ to show that

$$\frac{\partial}{\partial X_{i,j}} X^{-1} = -X^{-1} \frac{\partial X}{\partial X_{i,j}} X^{-1}, \quad i, j \in \mathbf{N}_m$$

Answer of Exercise 5 We first notice that

$$\frac{\partial}{\partial X_{i,j}} (XX^{-1}) = X \frac{\partial X^{-1}}{\partial X_{i,j}} + \frac{\partial X}{\partial X_{i,j}} X^{-1} = 0$$

by a previous exercise, from which the claim in the hint follows. This then results in

$$\frac{\partial X^{-1}}{\partial X_{i,j}} = -X^{-1} e_i e_j^T X^{-1}$$

and hence

$$\frac{\partial}{\partial X} a^T X^{-1} b = -a^T X^{-1} e_i e_j^T X^{-1} b = -e_i^T X^{-T} a b^T X^{-T} e_j$$

from which the result follows.

Exercise 6 Show that

$$\frac{\partial}{\partial X} \mathbf{tr} X^T A X = (A + A^T) X$$

where $X \in \mathbf{R}^{m \times n}$ and $A \in \mathbf{R}^{m \times m}$.

Answer of Exercise 6 It holds that

$$\begin{aligned} \frac{\partial}{\partial X_{i,j}} \mathbf{tr} X^T A X &= \mathbf{tr} \frac{\partial X^T}{\partial X_{i,j}} A X + \mathbf{tr} X^T A \frac{\partial X}{\partial X_{i,j}} \\ &= \mathbf{tr} e_j e_i^T A X + \mathbf{tr} X^T A e_i e_j^T \\ &= e_i^T A X e_j + e_j^T X^T A e_i = e_i^T (A + A^T) X e_j \end{aligned}$$

from which the result follows.

Convexity

Exercise 7 Let $f : \mathbf{R} \rightarrow \mathbf{R}$ and $g : \mathbf{R} \rightarrow \mathbf{R}$ be convex functions. Show that $f + g$ is also a convex function.

Answer of Exercise 7 For $\theta \in [0, 1]$ and any $x, y \in \text{dom}(f + g)$ we have $f(\theta x + (1 - \theta)y) + g(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y) + \theta g(x) + (1 - \theta)g(y) = \theta(f(x) + g(x)) + (1 - \theta)(f(y) + g(y))$, from which the result follows.

Exercise 8

1. Show that $f(x) = x \ln x$ with $\text{dom } f = \mathbf{R}_+$ is a convex function.
2. Show that $g(x) = \sum_{i=1}^n x_i \ln x_i$ with $\text{dom } g = \mathbf{R}_+^n$ is a convex function.
3. Derive the conjugate function of f above.

Answer of Exercise 8

1. We have $f'(x) = \ln x + 1$ and $f''(x) = 1/x > 0$ for $x \in \mathbf{R}_+$, and hence f is convex.
2. The function to $g(x) = yx - x \ln x$ to maximize has derivative $g'(x) = y - \ln x - 1$, which is zero for $x = e^{y-1}$. Moreover, $g''(x) = -1/x < 0$ for $x \in \mathbf{R}_+$, and hence e^{y-1} results in a maximum. Hence $f^*(y) = e^{y-1}(y - (y - 1)) = e^{y-1}$.

Exercise 9 Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ and define $\text{epi } f = \{(x, t) \mid x \in \text{dom } f, f(x) \leq t\}$. Show that f is a convex function if and only if $\text{epi } f$ is a convex set.

Answer of Exercise 9 Assume $\text{epi } f$ is a convex set. Then for all $(x, s), (y, t) \in \text{epi } f$ and $\theta \in [0, 1]$ it holds that $\theta(x, s) + (1 - \theta)(y, t) \in \text{epi } f$. Hence $f(\theta x + (1 - \theta)y) \leq \theta s + (1 - \theta)t$, and with $s = f(x)$ and $t = f(y)$ it follows that f is a convex function. Now assume that f is a convex function. Then for all $x, y \in \text{dom } f$ it holds that $f(\theta x + (1 - \theta)y) \leq \theta f(x) + (1 - \theta)f(y)$. Hence for all $s : f(x) \leq s$ and for all $t : f(y) \leq t$ it holds that $f(\theta x + (1 - \theta)y) \leq \theta s + (1 - \theta)t$, and hence $\text{epi } f$ is a convex set.

Exercise 10 Let $f : \mathbf{R}^n \rightarrow \mathbf{R}$ and let $g : \mathbf{R} \rightarrow \mathbf{R}$ be defined as $g(t) = f(x + vt)$. Show that f is a convex function if and only if g is a convex function for all $x, v \in \mathbf{R}^n$. Here $\text{dom } g = \{t \mid x + tv \in \text{dom } f\}$.

Answer of Exercise 10 The function g is convex if and only if for all $s, t \in \text{dom } g$ and for $\theta \in [0, 1]$ it holds that $f(x + v(\theta s + (1 - \theta)t)) \leq \theta f(x + vs) + (1 - \theta)f(x + vt)$. With $z = x + vs$ and $y = x + vt$ this can also be written as $f(\theta z + (1 - \theta)y) \leq \theta f(z) + (1 - \theta)f(y)$. Hence it follows that if f is convex, so is also g . Since any $z, y \in \text{dom } f$ can be written as above with e.g. $x = z$, $v = y - z$, $s = 0$ and $t = 1$ also the converse holds true.

Exercise 11 Show that $f(X) = -\ln \det X$ with $\text{dom } f = \mathbf{S}_{++}^n$ is a convex function. Hint: Consider $g(t) = -\ln \det(X + Vt)$, where $X \in \mathbf{S}_{++}^n$ and $V \in \mathbf{S}^n$.

Answer of Exercise 11 We have

$$\begin{aligned} g(t) &= -\ln \det(X + Vt) = -\ln \det(X^{1/2}(I + X^{-1/2}VX^{-1/2}t)X^{1/2}) \\ &= -\sum_{i=1}^n \ln(1 + \lambda_i t) - \ln \det X \end{aligned}$$

where $\lambda_i, i \in \mathbf{N}_n$ are the eigenvalues of $X^{-1/2}VX^{-1/2}$. Hence

$$g'(t) = -\sum_{i=1}^n \frac{\lambda_i}{1 + \lambda_i t}; \quad g''(t) = \sum_{i=1}^n \frac{\lambda_i^2}{1 + \lambda_i t} \geq 0$$

and therefore f is convex.